



Matrix normal PCA for interpretable dimension reduction and graphical noise modeling

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ABSTRACT

Principal component analysis (PCA) is one of the most widely used dimension reduction and multivariate statistical techniques. From a probabilistic perspective, PCA seeks a low-dimensional representation of data in the presence of independent identical Gaussian noise. Probabilistic PCA (PPCA) and its variants have been extensively studied for decades. Most of them assume the underlying noise follows a certain independent identical distribution. However, the noise in the real world is usually complicated and structured. To address this challenge, some variants of PCA for non-IID data have been proposed. However, most of the existing methods only assume that the noise is correlated in the feature space while there may exist two-way structured noise. To this end, we propose a powerful and intuitive PCA method (MN-PCA) through modeling the graphical noise by the matrix normal distribution, which enables us to explore the structure of noise in both the feature space and the sample space. MN-PCA obtains a low-rank representation of data and the structure of noise simultaneously. And it can be explained as approximating data over the generalized Mahalanobis distance. We first solve this model by a standard approach – maximizing the regularized likelihood – and then develop a novel algorithm that exploits the Wasserstein distance, which is more robust. Extensive experiments on various data demonstrate their effectiveness.

1. Introduction

Massive data emerge from diverse fields of science and engineering dramatically. For example, biologists can use microarray to detect the expression of thousands of genes at the same time; a single picture consists of hundreds, thousands, or even more pixels. Generally, those data are redundant and noisy. Directly extracting useful information from primitive data is often infeasible. Therefore, how to discover the compact and meaningful representation of high-dimensional data becomes a fundamental problem. Researchers have developed many powerful methods to address this problem from different perspectives. Among those methods, principal component analysis (PCA) is one of the most fundamental and widely used data mining and statistical techniques.

PCA obtains a low-dimensional representation for high-dimensional data in a L_2 -norm sense. As a consequence, PCA is sensitive to gross errors. To remove the effect of sparse gross errors, robust PCA (RPCA) [1, 2] has been proposed. Specifically, RPCA seeks to decompose the data matrix as a superposition of a low-rank matrix with a sparse

matrix. The sparse matrix captures the gross errors, enabling RPCA to recover the low-rank representation of data accurately. In the same spirit, researchers also proposed robust variants of nonnegative matrix factorization [3,4] by replacing the L_2 -norm with robust loss functions. Furthermore, Kang et al. [5] introduced a self-paced framework for PCA, which allowed for the adaptive reduction of the effects of noise and outliers.

To tackle the noise of data implicitly, regularized methods have been introduced. A common assumption is that the representations of two data points that are close should be close too. To preserve the local geometrical structure in the representation, graph regularizers are imposed. Shen et al. [6] unified various local manifold embedding methods and regularized the loss with graph embedding.

Probabilistic approaches are natural ways to account for different types of noise in data. Tipping and Bishop [7] first introduced the probabilistic PCA (PPCA), and showed that PCA could be derived from a Gaussian latent variable model with independent and identical distributed (IID) Gaussian noise. This probabilistic framework allows

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various extensions of PCA. For example, Bishop extended PCA by adopting a Bayesian method [8] (BPCA) that can automatically determine the number of principal components retained. Despite the Gaussian distribution, researchers have also introduced different distributions to handle different types of noise [9,10]. For example, Virta and Artemiou [10] introduced Poisson PCA for matrix count data. Most of those variants assume that the distribution of noise is IID.

However, the noise in the real world is usually complicated and structured. The IID assumption no longer holds. For example, data collected from different sources may contain heterogeneous noise. Zhang and Zhang [11] developed a Bayesian joint matrix decomposition method (BJMD) to model the heterogeneous noise of multi-view data by the Gaussian distribution. Subsequently, Zhang et al. [12] extended this work by developing an efficient distributed algorithm tailored for large-scale multi-view data with heterogeneous Gaussian noise. Some authors relaxed the identical assumption and introduced a mixture of distributions to model the complex noise [13]. Beyond the independent assumption, some researchers also explored the non-IID assumption in the sense of PCA [14,15]. Kalaitzis et al. [14] generalized PPCA by adding a fixed random effect to decompose the residual variance (RCA). RCA can be interpreted as the IID Gaussian noise with a linear offset in the feature space. Vaswani and Guo [15] studied the correlated-PCA problem where the noise is data-dependent. Barbier et al. [16] examined the influence of statistical dependencies in measurement noise on high-dimensional inference within the spiked matrix model of PCA, highlighting the necessity of utilizing both signal and noise structures to enhance signal estimation.

Most existing methods assume that the noise is only correlated in the feature space. However, the noise can be two-way correlated among features and among samples simultaneously. Take the gene expression data $Y \in \mathbb{R}^{n \times p}$ as an example, where n is the number of biological samples, and p is the number of genes (features). The noise of gene expression of samples may demonstrate different patterns under different biological conditions, while the noises of genes under the same biological processes are also correlated. There are only a few works that model the noise structure in feature and sample spaces simultaneously. Allen et al. [17] proposed the generalized PCA (GPCA) using the matrix normal distribution to model noise. GPCA finds the best low-rank approximation of data with respect to the predefined sample and feature precision matrices. GPCA works well for spatial-temporal data, but the predefined precision matrix can make it limited for general types of data.

To this end, we propose a powerful and intuitive PCA model (MN-PCA) to obtain the low-rank representation and the structure of the underlying noise at the same time. Unlike GPCA, MN-PCA infers the precision matrices of the matrix normal distribution from the observed data. The inference of the precision matrices of the matrix normal distribution turns to be that of two Gaussian graphical models, enabling us to explore the structure of the noise in both the feature space and the sample space. We first solve this model by a standard approach, i.e., maximizing the regularized likelihood, and then develop a novel algorithm that minimizes the Wasserstein distance between the distribution of the white matrix normal and that of the residual. We extensively compare the two algorithms and discuss their advantages and disadvantages. Extensive experiments on various data show their effectiveness. By considering the structure of the underlying noise, MN-PCA generally obtains a better low-rank representation, and the inferred structure of the noise is interpretable, revealing some interesting information that is ignored by other methods.

The contribution of this paper is threefold. First, we propose an intuitive framework MN-PCA that obtains the low-rank representation and discovers the noise structure in the sample and feature spaces at the same time; MN-PCA connects two important statistic topics, PCA and sparse precision matrix estimation. Second, we develop two effective algorithms from different perspectives; the algorithm based on minimizing the Wasserstein distance is more robust than maximizing

the regularized likelihood. Third, we provide theoretical analysis on the error bounds of the eigenvalues and eigenvectors when the noise follows a matrix normal distribution for conventional PCA. This suggests that the conventional PCA could have poor performance in terms of low-rank signal recovery when the noise is correlated, which justifies the rationale of MN-PCA. It may inspire future research on PCA for data with complicated noise.

2. Related work

In this section, we provide a brief review of two important statistical topics, PCA and sparse precision matrix estimation, as MN-PCA uniquely connects these domains.

2.1. PCA with non-IID assumption

PPCA [7] assumes that the distribution of noise is independent and identical distributed (IID) Gaussian distribution $Y = X + E$, where the data matrix Y can be decomposed into a low-rank matrix X and IID Gaussian noise matrix E . To account for general noise, Zhao et al. [13] relaxed the IID assumption and introduced the mixture of Gaussian noise to model the complex noise. Note that such noise is not identical but still independent. Inspired by the linear mixed model, Kalaitzis and Lawrence [14] proposed RCA to generalize PPCA by adding fixed effects

$$Y = XW^T + ZV^T + E, \quad (1)$$

where Z is a matrix of known covariates. The loading matrices W and V can be marginalized with the Gaussian isotropic priors:

$$\ln p(Y) = \sum_{j=1}^n \ln \mathcal{N}(Y_j | 0, XX^T + \Sigma), \quad (2)$$

where Y_j indicates the j th column of Y and $\Sigma = ZZ^T + \sigma^2 I$. Σ can be interpreted as an offset of the covariance in the feature space. There exists a line of research that studies PCA model for data-dependent noise [15,18,19]. For example, Gu and Shen [18] considered the problem that the noise is IID Gaussian but the factor is modeled by a zero-mean Gaussian process.

Utilizing the covariances in both the sample and feature spaces, Allen et al. [17] proposed a generalized PCA (GPCA):

$$Y = \sum_{i=1}^k d_i x_i w_i^T + E; E \sim \mathcal{MN}(\mathbf{0}, \Omega, \Sigma), \quad (3)$$

such that $X^T \Omega X = I$, $W^T \Sigma W = I$, and E follows the matrix normal distribution $\mathcal{MN}_{n,p}(M, \Omega, \Sigma)$. We note that Ω^{-1} and Σ^{-1} are predefined in GPCA. Hence, GPCA can be restrictive for general data.

To handle nonlinear data, KPCA extends PCA by adopting the kernel idea [20]. GPLVM makes use of the Gaussian process and is a nonlinear extension of PPCA [21]. However, those nonlinear methods are often difficult to explain.

In tensor data analysis, some researchers also made their efforts to handle complicated noise in real data [22,23]. For example, Ding and Cook [22] proposed dimension folding PCA (DFPCA) for matrix-valued data $Y = M + XZW^T + E$, where $X \in \mathbb{R}^{n \times k_1}$, $Z \in \mathbb{R}^{k_1 \times k_2}$, $W \in \mathbb{R}^{p \times k_2}$, and E follows the matrix normal distribution. The observed data are a set of matrices $\{Y_i\}_{i=1}^q$. Different from GPCA that the precision matrices are predefined, DFPCA estimates them from data by maximum likelihood estimation. But it requires the number of observed matrices $q > \max(n/p, p/n) + 1$ [24]. Hence, DFPCA is not applicable when there is only one observed matrix Y .

2.2. Gaussian graphical model

Gaussian graphical models (GGMs) have been proposed to understand the statistical relationship between variables of interest in the

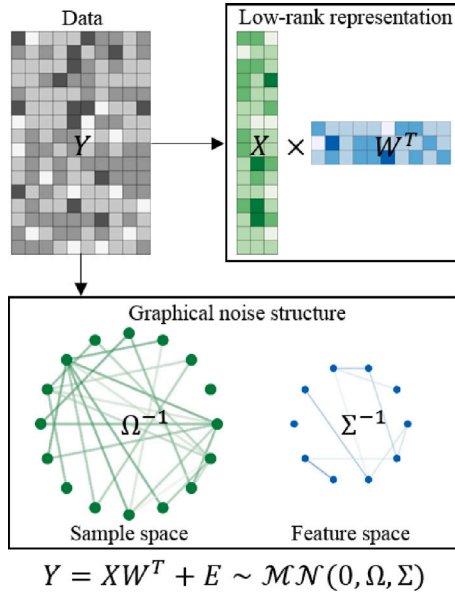


Fig. 1. Illustration of MN-PCA. MN-PCA models the low-rank representation and the structure of noise in both feature and sample spaces.

form of a graph. Those models use the precision matrix (the inverse of covariance) of the multivariate Gaussian to capture the conditional correlations between pairs of variables. Formally, suppose one has n multivariate normal observations of dimension p with covariance Σ . Let $\Theta = \Sigma^{-1}$ be the precision matrix, and S be the empirical covariance matrix; then the problem is to maximize the log-likelihood $\log |\Theta| - \text{tr}(S\Theta)$, where Θ is positive definite and $|\Theta|$ is the determinant of Θ .

To estimate a large precision matrix, the sparse assumption is made in literature: $\log |\Theta| - \text{tr}(S\Theta) - \rho \|\Theta\|_1$, where $\rho > 0$ controls the sparsity of Θ . This problem has been extensively studied. Popular algorithms include glasso [25], QUIC [26], and ACLIME [27].

3. Matrix normal PCA

3.1. Model construction

Most probabilistic methods assume the underlying noise E is IID. Though the IID assumption enjoys good theoretical properties, it is easily violated in real-world data. To model the underlying correlated noise (graphical noise), a naive approach is to assume that the noise follows a multivariate Gaussian distribution, i.e., $\text{vec}(E) \sim \mathcal{N}(\mathbf{0}, \Sigma)$. E consists of np variables. Hence, the corresponding covariance matrix Σ of size n^2p^2 is too huge.

Practitioners are often interested in the correlation relationships among samples and among features simultaneously. In the previous gene expression example, biologists are interested in the underlying structures of samples (e.g., disease subtypes) and features (gene modules). Hence, we assume that the noise is correlated in both the sample and feature spaces. We use a matrix normal distribution to model the among-sample and among-feature covariances of noise simultaneously. Due to its close relationship with PCA, we name it as MN-PCA (Fig. 1). Specifically, MN-PCA follows a similar generative model as PPCA, i.e., $Y = XW^T + E$. In addition, MN-PCA models the graphical noise by assuming $E \sim \mathcal{MN}_{n,p}(\mathbf{0}, \Omega, \Sigma)$, where $\Omega_{n \times n}$ and $\Sigma_{p \times p}$ are among-row and among-column variance, respectively. Note that the matrix normal is related to the multivariate normal distribution in the following way: $Y \sim \mathcal{MN}_{n,p}(M, \Omega, \Sigma)$, if and only if $\text{vec}(Y) \sim \mathcal{N}_{np}(\text{vec}(M), \Sigma \otimes \Omega)$, where $\otimes$ denotes the Kronecker product. Therefore, the density

function of $\mathcal{MN}_{n,p}(M, \Omega, \Sigma)$ has the form:

$$P(Y|M, \Omega, \Sigma) = \frac{\exp \left[-\frac{1}{2} \text{tr} \left(\Sigma^{-1} (Y - M)^T \Omega^{-1} (Y - M) \right) \right]}{(2\pi)^{\frac{np}{2}} |\Omega|^{\frac{n}{2}} |\Sigma|^{\frac{p}{2}}}. \quad (4)$$

Accordingly, the negative log-likelihood of MN-PCA is

$$\mathcal{L} = \frac{1}{2} \text{tr} \left[\Sigma^{-1} (Y - XW^T)^T \Omega^{-1} (Y - XW^T) \right] + \frac{p}{2} \log |\Omega| + \frac{n}{2} \log |\Sigma|. \quad (5)$$

To deal with large scale matrix and estimate precision matrix robustly, we impose sparse constraints on the precision matrices Ω^{-1} and Σ^{-1} by introducing the L_1 -norm regularization terms:

$$\mathcal{L} = \frac{1}{2} \text{tr} \left[\Sigma^{-1} (Y - XW^T)^T \Omega^{-1} (Y - XW^T) \right] + \frac{p}{2} \log |\Omega| + \frac{n}{2} \log |\Sigma| + n\lambda_1 \|\Omega^{-1}\|_1 + p\lambda_2 \|\Sigma^{-1}\|_1, \quad (6)$$

where λ_1, λ_2 control the sparsity of Ω^{-1} and Σ^{-1} , respectively.

We adopt a block coordinate descent strategy to maximize (6). In the following, we will refer to this algorithm as Maximum Regularized Likelihood (MRL). For fixed Ω and Σ , we employ an alternative least square approach (ALS) to update X and W

$$X = Y \Sigma^{-1} W (W^T \Sigma^{-1} W + \epsilon I)^{-1}, \quad (7)$$

$$W = Y^T \Omega^{-1} X (X^T \Omega^{-1} X + \epsilon I)^{-1}, \quad (8)$$

where ϵ is a small positive constant to avoid singularity. For fixed X and W , we optimize Ω and Σ iteratively as follows:

$$\begin{cases} \hat{\Omega} = \arg \min_{\Omega} \text{tr}(\Omega^{-1} S_1) + \log |\Omega| + \lambda_1 \|\Omega^{-1}\|_1, \\ \hat{\Sigma} = \arg \min_{\Sigma} \text{tr}(\Sigma^{-1} S_2) + \log |\Sigma| + \lambda_2 \|\Sigma^{-1}\|_1, \end{cases} \quad (9)$$

where $S_1 = \frac{1}{p} (Y - XW^T) \Sigma^{-1} (Y - XW^T)^T$ and $S_2 = \frac{1}{n} (Y - XW^T)^T \Omega^{-1} (Y - XW^T)$. In principle, we can solve (9) by any sparse precision matrix estimation algorithm. Here we adopt QUIC (QUadratic approximation of Inverse Covariance matrices) [26] as the basic solver for its efficiency (see Supplementary Material Section 3.1 for a comparison of QUIC and glasso [25]). The inference scheme is summarized in Algorithm 1.

Algorithm 1 MN-PCA

Input: data matrix Y , rank k

Output: X , W , Ω and Σ

- 1: Truncated k SVD of $Y = U_k \Sigma_k V_k$
 - 2: Initialize $X = U_k$ and $Y = \Sigma_k V_k$, $\Omega = I$, $\Sigma = I$
 - 3: **repeat**
 - 4: update Ω and Σ with (9)
 - 5: **repeat**
 - 6: update X with (7)
 - 7: update W with (8)
 - 8: **until** convergence
 - 9: **until** Change of the objective function value is small enough
-

3.2. Minimizing wasserstein distance

MRL is well-understood and easy to implement, yet it is still a point estimation and may get trapped in local minimums that are far from the global minimum. In the setting of MN-PCA, when the condition number of the among-row variance Ω or the among-column variance Σ is large, the data matrix Y can be significantly influenced by the noise E . To address this issue, we aim to minimize the discrepancy between the white matrix normal distribution and that of the residual instead of maximizing the likelihood function.

First, recall the original model $Y \sim \mathcal{MN}_{n,p}(M, \Omega, \Sigma)$ is equivalent to $QYR \sim \mathcal{MN}_{n,p}(QMR, I, I)$ where Q and R are both square, $Q^T Q =$

Ω^{-1} , $R^T R = \Sigma^{-1}$ and $M = XW^T$. It implies that the true factorization of Ω^{-1} and Σ^{-1} transform the residual matrix to IID Gaussian noise. Then we transform the objective function by defining a divergence $D(\cdot \| \cdot)$:

$$\min_{Q,R} D(Q(Y - M)R \| \mathcal{MN}_{n,p}(\mathbf{0}_{n \times p}, I_n, I_p)). \quad (10)$$

The Wasserstein distance [28] is a good metric of measuring the divergence between two distributions. If the Wasserstein distance is minimized to 0, the density functions of the two distributions are the same except a set with measure 0. Compared with MRL, minimizing the Wasserstein distance has the potential to find a solution closer to the global optimum. The definition of Wasserstein distance is $W_c(P_x, P_y) = \inf_T \mathbb{E}_{x \sim P_x} [c(x, T(x))]$, where $T : X \rightarrow Y$ is a measure preserving map. The Wasserstein distance is hard to compute due to the infimum operator. However, if both P_x and P_y are Gaussian distributions, their Wasserstein distance has a closed form solution. Given the two normal distributions $x = \mathcal{N}(m_1, \Sigma_1)$ and $y = \mathcal{N}(m_2, \Sigma_2)$ with the means $m_1, m_2 \in \mathbb{R}^p$ and the covariance $\Sigma_1, \Sigma_2 \in \mathbb{R}^{p \times p}$, their squared Wasserstein distance is defined as [29]:

$$W(x, y)_2^2 = \|m_1 - m_2\|_2^2 + \text{tr} \left(\Sigma_1 + \Sigma_2 - 2 \left(\Sigma_1^{\frac{1}{2}} \Sigma_2 \Sigma_1^{\frac{1}{2}} \right)^{\frac{1}{2}} \right). \quad (11)$$

We use E to denote the current residual, $E = Y - M$, $\tilde{E}$ to denote QER , and σ^2 to denote the variance of noise. To pursue sparsity, we introduce the L_1 -norm regularization as before. Note that the expectation of mean of E is zero, so we omit the computation of mean for both the norm and covariance in (11). When we update Q , the estimation of covariance among rows should be $\Sigma = \frac{1}{p} \tilde{E} \tilde{E}^T$, then the objective function turns to:

$$\min_Q \text{tr} \left(\Sigma - 2\sigma \Sigma^{\frac{1}{2}} \right) + \lambda_1 \|Q^T Q\|_1, \quad (12)$$

Since $\text{tr}(\Sigma) = \frac{1}{p} \text{tr}(\tilde{E} \tilde{E}^T) = \frac{1}{p} \|\tilde{E} \tilde{E}^T\|_2$ and $\text{tr}(\Sigma^{\frac{1}{2}}) = \frac{1}{\sqrt{p}} \text{tr}(\tilde{E} \tilde{E}^T)^{\frac{1}{2}} = \frac{1}{\sqrt{p}} \|\tilde{E}\|_*$, the objective function to update Q takes the form:

$$\min_Q \frac{1}{p} \|QER\|_2^2 - \frac{2\sigma}{\sqrt{p}} \|QER\|_* + \lambda_1 \|Q^T Q\|_1. \quad (13)$$

Similarly, the objective function to update R is:

$$\min_R \frac{1}{n} \|QER\|_2^2 - \frac{2\sigma}{\sqrt{n}} \|QER\|_* + \lambda_2 \|R^T R\|_1. \quad (14)$$

In (13), the coefficient ratio of terms $\|QER\|_2^2$ and $\|QER\|_*$ is $1/2\sigma\sqrt{p}$ and in (14), the ratio is $1/2\sigma\sqrt{n}$. If we update Q and R through Eqs. (13) and (14) alternatively, the optimization will not converge to the solution, since the coefficient ratios are not consistent. To address this issue, we unify the coefficients in Eqs. (13) and (14) with their geometric average $1/\sqrt{np}$ and $2\sigma/\sqrt[4]{np}$. Then the objective function is changed to:

$$\min_{Q,R} \frac{1}{\sqrt{np}} \|QER\|_2^2 - \frac{2\sigma}{\sqrt[4]{np}} \|QER\|_* + \lambda_1 \|Q^T Q\|_1 + \lambda_2 \|R^T R\|_1. \quad (15)$$

Not that if

$$Q^*, R^* = \arg \min_{Q,R} \frac{1}{\sqrt{np}} \|QER\|_2^2 - \frac{2}{\sqrt[4]{np}} \|QER\|_*, \quad (16)$$

then

$$\sqrt{\sigma} Q^*, \sqrt{\sigma} R^* = \arg \min_{Q,R} \frac{1}{\sqrt{np}} \|QER\|_2^2 - \frac{2\sigma}{\sqrt[4]{np}} \|QER\|_*. \quad (17)$$

Thus, there is no need to adjust the parameter σ . In the experiments, we find that the optimizing process converges faster with smaller σ . To keep the balance between Q and R , we normalize Q after each iteration, i.e., the average eigenvalues of Q and R are equal. Besides, the first term of the relaxed objective function (15) is quadratic to Q and R , which can be solved efficiently. To compute the gradient with

respect to Q and R through (15), we need to know the gradient of the nuclear norm $\|QER\|_*$. Since the nuclear norm is non-differentiable, we can only obtain the subgradient. If the SVD of QER is $U_0 \Sigma_0 V_0^T$, then a subderivative is $U_0 V_0^T$ [30]. Then the subgradients of Q and R through (15) is:

$$\nabla_Q = \frac{2}{\sqrt{np}} QER R^T E^T - \frac{2}{\sqrt[4]{np}} U_0 V_0^T R^T E^T + \lambda_1 \text{sign}(Q^T Q) Q, \quad (18)$$

$$\nabla_R = \frac{2}{\sqrt{np}} E^T Q^T QER - \frac{2}{\sqrt[4]{np}} E^T Q^T U_0 V_0^T + \lambda_2 \text{sign}(R^T R) R. \quad (19)$$

Then Q and R are updated by:

$$Q^+ = Q - \eta \nabla_Q, R^+ = R - \eta \nabla_R, \quad (20)$$

where η is the learning rate. The use of Wasserstein metric is inspired from generative adversarial network (GAN) [31], especially the Wasserstein-GAN [32]. Note that if we treat Q and R as the weight matrices of a layer of neural network, the data matrix as input and the truncated SVD as the activate function over the eigenvalue of Y , our MN-PCA model can be regarded as a single layer network. In Wasserstein GAN, the Wasserstein distance between the generated distribution and the target distribution is estimated by a discriminator, while we can directly compute the Wasserstein distance with a close form formula in MN-PCA. We implement the algorithm by Pytorch with the Adam optimizer.

If the likelihood function is maximized, the residual may not be Gaussian. In contrast, if the Wasserstein distance of two distributions is minimized, then the two distributions are the same almost everywhere. In MN-PCA, theoretically, we have

Proposition 1. Suppose E and R are fixed and $n > p$. Let $\text{row}\{QER\}$ denote the set of rows of the matrix QER , S is the set of n samples from the distribution $N(\mathbf{0}, I_p)$, H is a $n \times p$ matrix by stacking each item in S as a row in a certain order. Then $W(\text{row}\{QER\}, S)$ is minimized if and only if QER can be transformed to H_{p-k} by rearranging its rows, where $p-k$ is the rank of E and H_{p-k} is the truncated $p-k$ SVD of H .

Proposition 1 shows that under moderate assumptions, the minimum of Wasserstein distance corresponds to the global optimum of the problem. However, computing the Wasserstein distance exactly is prohibitive. We adopt the Wasserstein distance of two multivariate Gaussian in (11) instead. However, the real distribution of the residual is unknown for that the low-rank part is not exactly equal to the true signal during the optimization process. By utilizing (11), we actually approximated the distribution of residual with its mean vector and covariance matrix. Thus, the computation of Wasserstein distance is inaccurate. Since our model is over-parameterized, if we enforce the residual to be exactly IID Gaussian, the model can be overfitted. Thus, the inaccuracy in (11) makes it possible to adjust our model to different kinds of data. Moreover, the computation of the formula does not need sampling from the distribution of IID Gaussian, which is also intractable. We relegate all the proofs to the Supplementary Material Section 1.

Algorithm 2 MN-PCA-W2

Input: data matrix Y , rank k , λ_1 , λ_2

Output: X , $\Omega = (Q^T Q)^{-1}$ and $\Sigma = (R^T R)^{-1}$

- 1: Q and R are initialized as identity matrices
 - 2: **repeat**
 - 3: Truncated k SVD of $X = (QYR)_k$
 - 4: Set $E = Y - Q^{-1} X R^{-1}$
 - 5: Update Q and R with (20)
 - 6: Normalize $Q := \gamma Q$ where $\gamma := \text{tr}(R)/\text{tr}(Q)$
 - 7: **until** convergence
-

We summarized the computational complexity and convergence of the two algorithms in Supplementary Material Section 2.3. Empirically, the objective function value of MRL gets stable in less than ten rounds of iterations and MN-PCA-W2 usually needs more time than MRL.

3.3. Model interpretation

PCA aims at minimizing the reconstruction errors, measured by the sum of Euclidean distances between the data points Y_i and the reconstructed points $X_i W^T$. In particular, the objective function of PCA can be rewritten as:

$$\min \sum_{i=1}^n (Y_i - X_i W^T)(Y_i - X_i W^T)^T, \quad (21)$$

where X_i and Y_i indicate the i th row of the X and Y , respectively. Nevertheless, measuring the distance among points by Euclidean distance can be misleading, knowing that they are realizations of the anisotropic multivariate distribution. For example, as shown in Fig. 2a., there are three points x_1 (square), x_2 (cross mark) and x_3 (triangle) drawn from an anisotropic 2D normal distribution $\mathcal{N}(\mu, \Sigma)$ (the density is illustrated by the blue dots). x_2 is the center of the distribution. The Euclidean distance between x_1 and x_2 is equal to that between x_2 and x_3 . This is misleading. Because the density of the distribution near x_1 is much lower than that near x_3 . Considering the density of distribution, the distance x_1 should be farther from x_2 than x_3 . Hence, Euclidean distance does not reflect the underlying random process $\mathcal{N}(\mu, \Sigma)$. Mahalanobis distance addresses this problem by utilizing the information of covariance:

$$d(x_i, x_j) = (x_i - x_j)^T \Sigma^{-1} (x_i - x_j), \quad (22)$$

which is equivalent to deform the anisotropic Gaussian to an isotropic Gaussian and then compute the Euclidean distance (Fig. 2b). MN-PCA can be regarded as minimizing the reconstruction error over the generalized Mahalanobis distance. When $\Omega = I$, the reconstruction error of MN-PCA has the Mahalanobis distance form:

$$\min \sum_{i=1}^n (Y_i - X_i W^T) \Sigma^{-1} (Y_i - X_i W^T)^T. \quad (23)$$

We have a similar result when $\Sigma = I$. Thus, we can consider MN-PCA as a minimization of the reconstruction error over the generalized Mahalanobis distance. MN-PCA transforms the matrix normal noise to isotropic noise and then applies PCA to the transformed data. The following theorem explains the rationale of MN-PCA:

Proposition 2. Suppose $Y = M + E$, where $M \in \mathbb{R}^{n \times p}$ is the low-rank structure, and $E \sim \mathcal{MN}(\mathbf{0}, \Omega, \Sigma)$ is the matrix normal noise. Assume that Y is already centered for simplicity. For $\sigma > 0$, if $\text{tr}(\Omega) = n\sigma$ and $\text{tr}(\Sigma) = p\sigma$ (the assumption makes the scales of IID noise $\mathcal{N}(0, \sigma^2)$ and matrix normal noise $\mathcal{MN}(\mathbf{0}, \Omega, \Sigma)$ comparable), then the following inequalities hold:

$$\left| \sigma_i \left(\frac{1}{n} \mathbb{E}(Y^T Y) \right) - \sigma_i \left(\frac{1}{n} M^T M \right) \right| \leq \sigma \sigma_1(\Sigma), \quad (24)$$

$$\left| \sigma_i \left(\frac{1}{p} \mathbb{E}(Y Y^T) \right) - \sigma_i \left(\frac{1}{p} M M^T \right) \right| \leq \sigma \sigma_1(\Omega), \quad (25)$$

where $\sigma_i(A)$ is the i th largest eigenvalue of the matrix A .

The inequalities (24) and (25) quantify the difference of eigenvalues between the expectation of sample covariance and the noise-free covariance. In (24), the upper bound $\sigma \sigma_1(\Sigma)$ is the tightest when the noise is IID, which implies that PCA is more suitable for IID noise and may be inaccurate when the noise is non-IID. The same argument applies to (25).

Moreover, considering that Y is the sum of a low-rank signal and a random matrix normal matrix, the upper bounds in (24) and (25) of the eigenvalue difference can be further refined. We use the theorems in [33] to refine the bounds between the eigenvalues of the signal M and the observed matrix Y . The following Propositions 2 and 3 are direct applications.

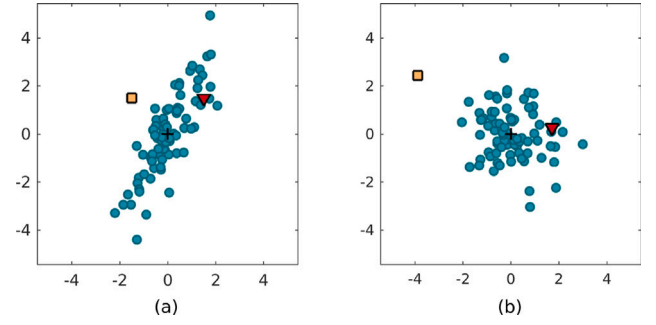


Fig. 2. Illustration of the Mahalanobis distance in a 2D data. The blue dots are drawn from a 2D Gaussian distribution. The cross mark indicates the center of the distribution.

Proposition 3. Suppose $Y = M + E$, where M is the low-rank signal of rank r and $E \sim \mathcal{MN}(0, \Omega, \Sigma)$. Then for any $t > 0$, $1 \leq i \leq r$,

$$\mathbb{P}(\sigma_i(Y) \geq \sigma_i(M) - t) \geq 1 - 4 \times 9^i \exp\left(-\frac{\beta t^2}{16}\right), \quad (26)$$

where $\beta := \frac{1}{2\sigma_1(\Omega)\sigma_1(\Sigma)}$. The upper bound of σ_i is characterized as

$$\mathbb{P}\left(\sigma_i(Y) \leq \sigma_i(M) + t\sqrt{r} + \frac{2\sqrt{i}\|E\|^2}{\sigma_i(Y)}\right) \geq 1 - 4 \times 9^{2r} \exp\left(-r\frac{\beta t^2}{16}\right), \quad (27)$$

where $\|E\| := \sigma(E)$ is the spectral norm.

The condition number of matrix A is defined as $c = \sigma_{\max}(A)/\sigma_{\min}(A) \geq 1$, where $\sigma_{\max}(A)$ and $\sigma_{\min}(A)$ are the largest and smallest singular values of matrix A , respectively. When $\text{tr}(\Omega) = n\sigma$ and $\text{tr}(\Sigma) = p\sigma$ are fixed, $\sigma_1(\Omega)$ and $\sigma_1(\Sigma)$ increase with the growing of their condition numbers c_1 and c_2 , respectively. Hence, the right hand side of (26) decreases with the increase of c_1 and c_2 , implying that the lower bound is tighter when the noise is isotropic. The same argument also applies to the right hand side of (26) since $\|E\|$ also grows along with the increase of c_1 and c_2 . The sine of the angle between the first eigenvectors is also dependent on β .

Proposition 4. Suppose $Y = M + E$, where M is the low-rank signal of rank r and $E \sim \mathcal{MN}(0, \Omega, \Sigma)$. Then, for any $t > 0$,

$$\begin{aligned} \mathbb{P}\left(\sin \angle(v_1(M), v_1(Y)) \leq 2^{\frac{5}{2}} \cdot \frac{t\sqrt{2}\sigma_1(M) + \|E\|\delta + \|E\|^2}{\sigma_1(M)\delta}\right) \\ \geq 1 - 108 \exp\left(\frac{-\beta\delta^2}{64}\right) - 4 \times 9^{2r} \exp\left(\frac{-r\beta t^2}{16}\right), \end{aligned} \quad (28)$$

where $\delta := \sigma_1(M) - \sigma_2(M)$ is the difference between the largest eigenvalue and the second largest eigenvalue of the signal matrix.

Again, the upper bound is tighter when the noise is isotropic when $\text{tr}(\Omega) = n\sigma$ and $\text{tr}(\Sigma) = p\sigma$ are fixed.

3.4. Tuning parameters selection

Given the desired dimension k , we should select the tuning parameters λ_1 and λ_2 to balance the trade-off of the log-likelihood value and the sparseness of the inferred precision matrices. While carrying a 2D grid search of λ_1 and λ_2 is computational expensive, we use the Bayesian Information Criterion (BIC) to select λ_1 and λ_2 separately. The BIC for the L_1 -norm regularized precision matrix for a fixed value of λ_1 is given in [34]:

$$\text{BIC}(\lambda_1) = -\log |\Omega^{-1}(\lambda_1)| + \text{tr}(S_1 \Omega^{-1}(\lambda_1)) + t_1 \frac{\log p}{p}, \quad (29)$$

where $\Omega^{-1}(\lambda_1)$ is inferred by the MRL algorithm. X and W are given by truncated k SVD and $\Sigma^{-1} = I$. t_1 is the number of non-zero entries in the upper diagonal portion of $\Omega^{-1}(\lambda_1)$.

Table 1

Comparison of PSNR on the small synthetic data sets.

c	PCA	CopulaPCA	GPCA	MN-PCA	MN-PCA-w2
8	15.71(0.24)	14.64(0.17)	16.55(0.27)	16.79(0.19)	16.64(0.17)
16	14.15(0.72)	13.92(0.32)	15.51(0.43)	16.34(0.28)	16.28(0.23)
32	9.21(1.48)	12.77(0.62)	11.99(1.99)	15.64(0.36)	13.12(3.55)
64	5.62(1.24)	10.31(1.92)	5.90(2.07)	11.07(4.16)	11.36(3.34)
96	3.09(1.49)	9.04(2.01)	2.61(1.31)	6.94(3.71)	10.37(3.02)
128	1.52(1.24)	7.70(2.12)	2.63(0.93)	4.32(1.09)	9.58(3.40)
160	0.43(1.14)	6.90(2.03)	0.95(1.90)	2.83(1.92)	9.33(3.19)
192	-0.42(1.18)	6.23(1.93)	0.46(1.91)	1.78(1.94)	9.62(3.24)
224	-1.14(1.24)	5.68(1.90)	-0.80(1.11)	0.54(1.80)	8.99(3.28)

Table 2

Comparison of NMI (%) on the small synthetic data sets.

c	PCA	CopulaPCA	GPCA	MN-PCA	MN-PCA-w2
8	98.93(1.26)	97.78(1.31)	99.15(1.22)	99.31(0.89)	98.22(1.69)
16	98.19(1.93)	98.27(1.08)	98.84(1.68)	99.29(0.91)	99.30(0.90)
32	73.87(12.75)	95.52(2.49)	86.68(11.33)	97.94(1.75)	82.88(18.82)
64	40.33(12.43)	84.30(11.81)	42.47(20.22)	70.00(34.42)	75.40(19.51)
96	17.89(14.10)	78.85(10.77)	13.79(13.94)	43.42(36.95)	70.16(18.95)
128	7.33(6.33)	68.67(13.38)	18.11(15.17)	25.62(27.49)	58.78(34.88)
160	3.68(2.21)	64.82(13.95)	7.78(16.43)	18.13(25.01)	59.25(34.74)
192	2.68(1.41)	62.11(13.52)	7.45(13.90)	17.39(25.24)	68.71(28.93)
224	2.08(1.07)	58.34(13.17)	1.81(0.92)	13.00(19.45)	63.86(27.14)

As suggested in [35], we set the largest value of the candidate λ_1 as $\lambda_{\max} = \max(\max(S_1 - I)_{ij} - \min(S_1 - I)_{ij})_{ij}$. We use a grid search to find λ_1 ($0.1\lambda_{\max} < \lambda_1 < \lambda_{\max}$) that minimizes the BIC. λ_2 is chosen by the same procedure. For MN-PCA-w2, we empirically set $\lambda_1 = \lambda_2 = 0.05$ in all the experiments see Figure S2 in the Supplementary Material for the sensitivity analysis.

4. Experimental results

In this section, we demonstrate the effectiveness of MN-PCA on both synthetic data and real-world data. Initially, we assess the low-rank recovery performance of MN-PCA by comparing it with PCA and its variants on both small-scale and large-scale synthetic datasets. Subsequently, we conduct a comparison between MN-PCA and the QUIC algorithm, illustrating that disregarding the low-rank structure leads to inaccurate precision matrix estimation. We further conduct extensive experiments on various real-world data. All experiments are performed on a desktop computer with a 2 GHz Intel Xeon E5-2683v3 CPU, a GTX 1080 GPU card and 16 GB memory. The code is available at <https://github.com/zhanglabtools/MN-PCA>.

4.1. Synthetic experiments

4.1.1. Small-scale synthetic data

We construct the toy data with $Y = M + E$, where $Y \in \mathbb{R}^{n \times p}$ is the observed data matrix, M is the low-rank signal and E is the matrix normal noise. Each row of M are drawn from k different centroids which are row vectors of length p . We then add the matrix normal noise $E \sim \mathcal{MN}(\mathbf{0}, \Omega, \Sigma)$ to M and use the sparsity parameters α_1 and condition number c_1 to control the generated Ω^{-1} randomly. In particular, the sparsity parameter indicates the proportion of nonzero entries in Ω^{-1} , $\alpha_1 = \#\{\Omega^{-1} \neq 0\}/n^2$. Note that the larger the condition number is, the violation of the IID assumption is more serious. Similarly, we generate Σ^{-1} with parameters α_2 and c_2 . We always set $c_1 = c_2 = c$, $\alpha_1 = \alpha_2 = \alpha$ for simplicity in the following experiments. Note that $c = 1$ and $\alpha = 0$ correspond to the IID situation.

We generate data with $n = 300$, $p = 200$, $k = 2$, $\alpha = 10^{-2}$. M consists of 300 samples drawn evenly from three different centroids. Specifically, the first and last 20 entries of centroid 1 equal one; the first and last 20 entries of centroid 2 equal minus one; the first 20 entries of centroid 3 equal one and the last 20 entries equal minus

one. To investigate the effects of the graphical noise, we vary the condition number c from 8 to 224. We generate 10 synthetic data for each parameter combination. The L_1 -norm penalty parameters λ_1 , λ_2 of the maximum likelihood are selected by the aforementioned procedure.

We investigate the effect of graphical noise on the performance of PCA, CopulaPCA [19], GPCA [17] with inverse smoothing precision matrices and MN-PCA measured in terms of low-rank matrix recovery. As aforementioned, PCA can be regarded as under IID assumption. In contrast, CopulaPCA and GPCA are two PCA variants specifically designed to accommodate non-IID assumptions. Specifically, let the estimated low-rank matrix be $\hat{M}$, and the true low-rank matrix be M^* . We calculate the peak signal to noise ratio (PSNR, Table 1) and normalized mutual information (NMI, Table 2) of k-means clustering on the projected data. It is worth noting that PSNR is a direct measure of the low-rank recovery, while NMI measures the low-rank recovery by clustering indirectly. The performance of all the methods tends to decrease as the condition numbers increase. PCA works well when the noise is close to an IID Gaussian ($c = 8$ and 16), which is aligned with our theoretical results in Propositions 1–3. GPCA consistently outperforms PCA because it mitigates the effect of the correlated noise by the predefined precision matrices, but it results in poor performance when the condition number is large. CopulaPCA incorporated with L_1 regularizers can potentially perform better in terms of NMI when the condition number of noise is large (Table 2, $c = 128, 160$), but its performance in PSNR is always inferior to the MN-PCA-W2. The superior performance of MN-PCA and MN-PCA-w2 over the other methods emphasizes the significance of learning the precision matrix from the data. In addition, MN-PCA-w2 is more robust and outperforms MN-PCA when the condition numbers are sufficiently large.

PCA, CopulaPCA and GPCA cannot infer the noise structure from the data. To facilitate the comparison in terms of precision matrix estimation, we apply QUIC to among-row and among-column covariances, respectively. Since there are many small values in the true precision matrices, we retain the top 150 non-diagonal entries (by absolute value) and summarize the performance by positive rate (TPR) and predictive positive value (PPV).

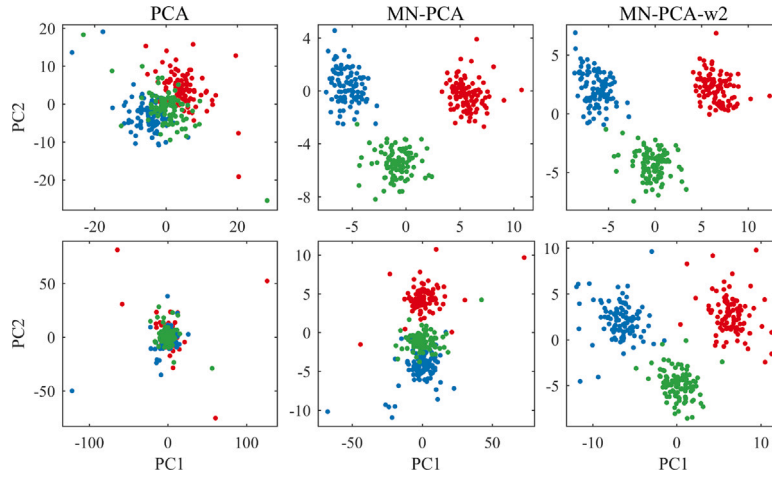
$$\text{TPR}(\Omega) = \frac{\#\{\Omega_{ij} \neq 0 \mid \Omega_{ij} \neq 0\}}{\#\{\Omega_{ij} \neq 0\}}, \text{PPV}(\Omega) = \frac{\#\{\Omega_{ij} \neq 0 \ \& \ \Omega_{ij} \neq 0\}}{\#\{\Omega_{ij} \neq 0\}}. \quad (30)$$

Numerical results from Table 3 reveal some interesting conclusions. (1) MN-PCA outperforms QUIC and MN-PCA-w2. (2) The TPR of all

Table 3

Performance of precision matrix estimation on the small synthetic data sets.

c	TPR(Ω^{-1})			PPV(Ω^{-1})		
	QUIC	MN-PCA	MN-PCA-w2	QUIC	MN-PCA	MN-PCA-w2
8	0.17(0.05)	0.35(0.08)	0.12(0.03)	0.58(0.12)	0.75(0.12)	0.19(0.04)
16	0.21(0.06)	0.45(0.11)	0.18(0.03)	0.33(0.08)	0.73(0.16)	0.26(0.04)
32	0.25(0.06)	0.45(0.10)	0.22(0.04)	0.33(0.05)	0.80(0.15)	0.31(0.03)
64	0.27(0.05)	0.38(0.09)	0.26(0.03)	0.34(0.05)	0.83(0.13)	0.34(0.03)
96	0.28(0.05)	0.39(0.09)	0.29(0.03)	0.35(0.05)	0.72(0.18)	0.37(0.03)
128	0.30(0.05)	0.38(0.10)	0.30(0.03)	0.37(0.04)	0.75(0.16)	0.38(0.02)
160	0.29(0.05)	0.34(0.09)	0.32(0.04)	0.37(0.04)	0.78(0.15)	0.39(0.03)
192	0.31(0.05)	0.33(0.09)	0.32(0.04)	0.38(0.04)	0.76(0.16)	0.39(0.03)
224	0.30(0.06)	0.31(0.09)	0.32(0.04)	0.38(0.05)	0.74(0.17)	0.39(0.03)
c	TPR(Σ^{-1})			PPV(Σ^{-1})		
	QUIC	MN-PCA	MN-PCA-w2	QUIC	MN-PCA	MN-PCA-w2
8	0.09(0.02)	0.22(0.05)	0.17(0.03)	0.12(0.03)	0.94(0.05)	0.20(0.02)
16	0.11(0.03)	0.30(0.06)	0.21(0.05)	0.14(0.03)	0.84(0.10)	0.24(0.04)
32	0.13(0.03)	0.28(0.07)	0.24(0.05)	0.16(0.03)	0.84(0.11)	0.26(0.03)
64	0.14(0.03)	0.19(0.06)	0.26(0.06)	0.18(0.03)	0.82(0.13)	0.28(0.04)
96	0.15(0.03)	0.22(0.06)	0.28(0.08)	0.18(0.03)	0.65(0.14)	0.29(0.05)
128	0.15(0.03)	0.19(0.05)	0.28(0.07)	0.18(0.03)	0.68(0.14)	0.28(0.05)
160	0.15(0.03)	0.16(0.05)	0.28(0.08)	0.18(0.03)	0.70(0.13)	0.28(0.05)
192	0.14(0.04)	0.16(0.04)	0.30(0.07)	0.18(0.04)	0.72(0.16)	0.30(0.05)
224	0.15(0.03)	0.14(0.05)	0.30(0.07)	0.18(0.03)	0.65(0.16)	0.30(0.05)

**Fig. 3.** Illustration of PCA, MN-PCA and MN-PCA-w2 on the small synthetic data. Top: $c = 32$. Bottom: $c = 224$.

methods is relatively low as there may exist many small values in the true precision matrices. The BIC criterion is prone to choose sparser models, as discussed in [36]. (3) The TPR of QUIC increases along with the increase of the condition numbers, because it is easier for QUIC to find true interactions when the effect of the matrix normal noise is stronger. (4) On the other hand, the TPR of the proposed methods increases first and then decreases with the condition numbers increase. When the effect of the matrix normal noise is small, it is difficult to capture the structure of the precision matrices. However, if the effect of the noise is too strong, the estimated low-rank matrix may mislead the estimation of precision matrices. (5) The PPV of MN-PCA is significantly higher than that of QUIC. Therefore, the FDR ($\text{FDR} = 1 - \text{PPV}$) of the proposed methods is acceptable when the condition number is mild. MN-PCA may help scientists to discover the underlying statistical correlations in the noise, which are ignored by PCA and most of its variants.

We illustrate the difference between MN-PCA and PCA on data generated with mild and large c , respectively. Fig. 3 shows that both MN-PCA and MN-PCA-w2 have clear clusters when the condition number is mild ($c = 32$). PCA has inferior performance when the condition number is large ($c = 224$, bottom panel). In addition, the projection of MN-PCA is more dispersed than that of MN-PCA-w2, suggesting that minimizing the Wasserstein distance is more accurate when the condition number is large.

4.1.2. Synthetic data with general noise and dense noise structure

The distribution of noise in the real-world data may be apart from matrix normal distribution. To evaluate the performance of MN-PCA when the noise is of general types, we first sample IID noise $\bar{E}$ from a distribution or a mixture of distributions. The transformed correlated noise matrix $E := Q\bar{E}R$, where Q and R are two symmetric linear transformations such that $Q^T Q = \Omega$ and $R^T R = \Sigma$. Specifically, we generate $\bar{E}$ from (1) Laplace distribution $L(0, 1)$, (2) uniform distribution $U(-1, 1)$, (3) Mixture1: Gaussian mixture distribution with 30% $N(0, 0.2)$, 30% $N(0, 0.5)$ and 40% $N(0, 1)$, (4) Mixture2: Gaussian mixture distribution with 40% $N(0, 0.5)$, 50% $N(0, 1)$ and 10% $N(0, 3)$, and (5) Mixture3: a mixture distribution with 45% $N(0, 0.5)$, 45% $N(0, 1)$ and 10% $U(-1, 1)$. We generate the synthetic data sets with $c = 32$ and $c = 64$, respectively.

The experimental result is consistent with that on the small synthetic data sets. Tables 4 and S3 show that the proposed methods outperform PCA and GPCA in all scenarios, suggesting that MN-PCA can work well even when the model is misspecified. CopulaPCA has a better performance when there exists gross noise (Mixture2). Compared with MN-PCA, CopulaPCA usually has higher NMI but lower PSNR, showing that it might get better clustering results but does not faithfully recover the low-rank signal. Since the precision matrices of GPCA is predefined, GPCA tends to be more sensitive than MN-PCA when the distribution of the noise violates the matrix normal distribution assumption.

Table 4

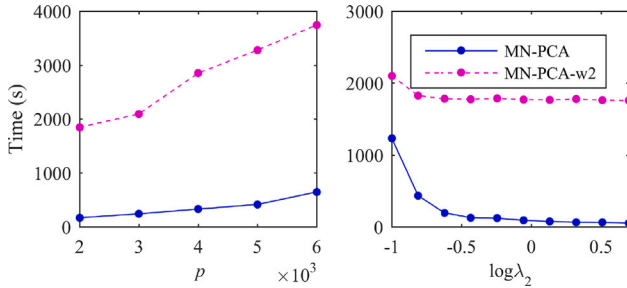
Comparison of PSNR on the synthetic data sets with general noise types.

Noise type	PCA	CopulaPCA	GPCA	MN-PCA	MN-PCA-w2
covariance matrix condition numbers $c_1 = c_2 = 32$					
Laplace	9.06(1.35)	13.11(0.59)	10.73(2.57)	15.66(0.30)	13.66(0.99)
Uniform	18.04(1.02)	16.38(0.48)	2.60(6.20)	20.52(0.28)	20.15(0.40)
Mixture1	14.02(1.03)	14.79(0.56)	15.94(0.86)	18.07(0.36)	17.17(0.50)
Mixture2	9.20(1.36)	12.81(0.66)	11.42(2.09)	15.71(0.34)	14.06(0.75)
Mixture3	15.44(2.00)	15.47(0.37)	17.80(0.68)	19.17(0.28)	18.61(0.38)
covariance matrix condition numbers $c_1 = c_2 = 64$					
Laplace	5.29(1.57)	11.26(1.27)	5.09(1.53)	12.99(3.46)	5.82(1.46)
Uniform	12.11(3.20)	15.35(0.72)	10.50(5.90)	19.74(0.52)	18.91(0.70)
Mixture1	8.30(1.04)	13.32(1.07)	8.97(2.09)	16.23(3.23)	13.19(3.00)
Mixture2	5.96(1.09)	10.55(1.64)	5.73(1.24)	8.99(3.91)	6.23(1.22)
Mixture3	9.20(2.30)	14.11(0.77)	11.13(3.65)	16.70(3.67)	16.15(3.25)

Table 5

Comparison of PSNR on the synthetic datasets with dense noise structure.

θ	PCA	CopulaPCA	GPCA	MN-PCA	MN-PCA-w2
0.3	19.60(0.15)	19.29(0.37)	19.68(0.16)	19.75(0.16)	22.32(2.12)
0.6	17.40(0.24)	17.20(0.19)	17.49(0.25)	17.56(0.25)	24.09(3.19)
0.9	14.28(0.46)	15.17(0.35)	14.44(0.47)	14.60(0.48)	19.76(4.32)

**Fig. 4.** Running time of MN-PCA and MN-PCA-w2 on the synthetic data of $n = 1000$, $c = 192$. Left: time versus data size. Right: time versus $\log \lambda_2$ with $p = 2000$ and $\lambda_1 = 0.5$.

To further illustrate the effectiveness of MN-PCA, we construct a set of synthetic data sets following the procedure described in GPCA [17]. The temporal and spatial covariances $\Sigma_{ij} = \theta^{|i-j|}$ and $\Omega = \theta^{d_{ij}}$, where d_{ij} is the Chebyshev distance between variables. $0 \leq \theta \leq 1$ controls the amount of spatio-temporal correlation in the data. We note that the precision matrices are dense, which violates the assumption of MN-PCA. Table 5 and S4 report the PSNR and root of the mean square error (RMSE), respectively (NMI is not available on this data). The result is consistent with that in [17]. GPCA slightly outperforms PCA. CopulaPCA underperforms PCA when $\theta \leq 0.6$. In addition, MN-PCA-w2 obtains significantly better results than the other methods, demonstrating the superiority of minimizing the Wasserstein distance when the model is misspecified.

4.1.3. Large-scale synthetic data

We further investigate the performance of MN-PCA on the large-scale data. We generate data with $n = 1000$, $k = 2$, $c = 196$, $\alpha_1 = 10^{-2}$, $\alpha_2 = 10^{-3}$ and vary p from 2000 to 6000. The first 10% features of centroid 1 equal one; the last 10% features of centroid 2 equal one; the first 10% features of centroid 3 equal one and the last 10% features equal minus one. Tables 6 and Tables 7 show the performance of low-rank recovery performance (the results are the average of 5 runs). PCA fails to produce meaningful projection because the NMI is close to zero. MN-PCA-w2 significantly outperforms MN-PCA in terms of PSNR and RMSE. The projection of MN-PCA is more dispersed and thus may have a better performance in clustering, but MN-PCA-w2 tends to recover the low-rank matrix more accurately (higher PSNR). The same argument applies to CopulaPCA.

Fig. 4 show the running time of MN-PCA and MN-PCA-w2. MN-PCA is much faster than MN-PCA-w2 (left panel). The running time of MN-PCA drops down rapidly with the increase of λ_2 (right panel). Because a large λ_2 encourages the sparsity of the estimated precision matrix, allowing us to use the Cholesky decomposition on each block (see Supplementary Material Section 2.3). The running time of MN-PCA-w2 does not change a lot with the increase of λ_2 , because its implementation does not fully exploit the sparsity of the estimated precision matrices.

4.1.4. Swiss roll data

We then compare our methods with PCA on the Swiss roll data (Fig. 5, first panel), which is generated by rolling two-dimensional points on a plane into three-dimensional space. Existing algorithms such as SDE [37] considering the local geometry structure can unfold the roll to 2D space successfully. Fig. 5 shows that the two clusters in the projection of PCA are mixed, while the two proposed methods separates the two rolls clearly. Despite that MN-PCA does not consider the local geometry or geodesic distance explicitly, the precision matrices may reflect the local structure to some extent and allow for a better representation.

4.2. Real-world experiments

In this section, we show the effectiveness of MN-PCA and compare it with PCA as well as three other competing methods on various real-world data, including GPLVM, CopulaPCA, kernel PCA (KPCA) and GPCA. Since how to choose an appropriate kernel for the given data is still unknown, we apply KPCA with different kernels (i.e., linear and polynomial) and recommended parameters. GPCA is the most competing approach [17]. However, it requires the precision matrix Ω^{-1} and Σ^{-1} to be predefined. Two empirical ways have been suggested to set it: (1) the inverse smoothing matrix, which is computed by the Laplacian matrix L ; (2) the standard exponential smoothing matrix S , which is based on the distance matrix. Finally, there are four different settings for GPCA, denoted as LL, LS, SL, and SS, respectively. GPCA is designed for spatial-temporal data and might result in ill-conditioned precision matrices for general types of data. We normalize the predefined precision matrices by $\frac{A + \min(0, -\sigma_{\min}(A))I}{\sigma_{\max}(A) + \min(0, -\sigma_{\min}(A))}$, where $A = L, S$. The statistics of the 17 used data sets for evaluation are summarized in Table S7.

All methods require an approximation of the rank of the data matrix. We adopt the bi-cross validation method [38]. Specifically, we first

Table 6

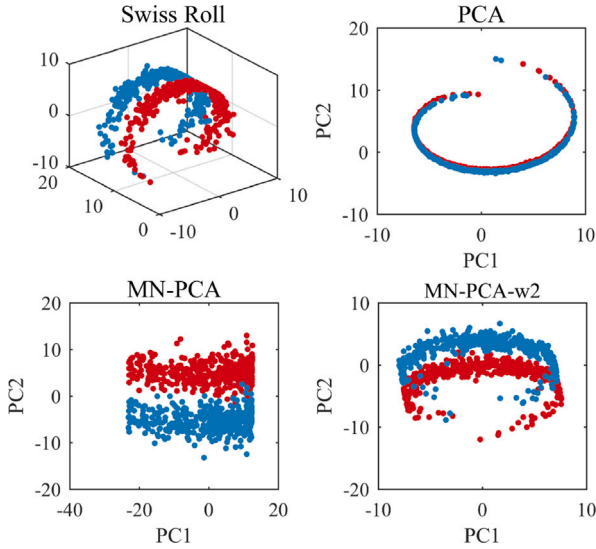
Comparison of NMI (%) on the large synthetic data sets.

p	PCA	CopulaPCA	GPCA	MN-PCA	MN-PCA-w2
2000	4.32(7.86)	76.75(3.03)	37.87(21.54)	98.80(1.18)	96.70(2.17)
3000	4.88(7.34)	79.00(3.63)	22.02(27.74)	85.13(9.73)	96.75(1.80)
4000	0.83(0.09)	69.22(1.35)	10.88(10.34)	90.75(11.66)	73.47(7.52)
5000	1.42(0.44)	71.06(4.79)	18.37(20.88)	82.98(11.91)	47.62(1.57)
6000	1.57(0.53)	71.35(6.92)	1.34(0.36)	59.43(32.38)	44.40(1.65)

Table 7

Comparison of PSNR on the large synthetic data sets.

p	PCA	CopulaPCA	GPCA	MN-PCA	MN-PCA-w2
2000	4.53(0.50)	9.97(3.03)	5.79(1.38)	20.06(1.19)	20.99(0.17)
3000	4.79(0.46)	10.02(3.63)	4.72(1.42)	7.49(0.65)	20.59(0.23)
4000	4.10(0.51)	9.34(1.35)	3.96(0.60)	12.51(5.74)	14.96(0.97)
5000	4.47(0.25)	9.20(4.79)	3.94(1.01)	7.76(0.38)	10.87(0.22)
6000	3.71(0.66)	8.84(6.92)	3.26(0.29)	5.60(2.30)	10.69(0.20)

**Fig. 5.** Illustration of PCA and MN-PCA on the Swiss roll data. From top left to bottom right: the 3D scatter plot of the Swiss roll data; the projection of PCA; the projection of MN-PCA; the projection of MN-PCA-w2.

approximate the data matrix by the truncated SVD $\hat{Y}_k$, where $\hat{Y}_k = \sum_{j=1}^k \hat{d}_j \hat{u}_j \hat{v}_j^T$. The proportion of variance explained is defined as $R_i = \sum_{j=1}^i \hat{d}_j^2 / \sum_{j=1}^K \hat{d}_j^2$. Note that R_i is between 0 and 1 and grows with i . We assume that the redundant components should not contribute much to the total variance. The final rank estimation is the smallest integer k , which satisfies $R_k > \tau$, $\tau > 0$ is the thresholding proportion value (e.g. 0.8) and $1 \leq k \leq K$.

In the following experiments, we set $K = 10$ and $\tau = 0.8$ to choose the rank for dimension reduction. If the number of features are smaller than 15, we simply set $k = 2$. Then we use a Least-squares SVM (LSSVM) [39] with a Gaussian kernel as a benchmark classifier to evaluate the quality of the low-rank representation. To facilitate a fair comparison, we choose the hyperparameter of LSSVM by 5-fold cross validation. The accuracy is the average of 10 runs.

Table 8 shows the cross-validated classification accuracies of these methods on the real-world data sets. We could find the following observations. (1) Our methods have competitive or superior performance compared with PCA in most cases, suggesting that MN-PCA could obtain more informative representation in some real applications. (2) GPCA with proper regularization on the predefined precision matrices generally works well. GPCA is known to work well on spatial-temporal data. However, the predefined precision matrices weaken its adaptivity to general types of data. For example, 3sources is a text dataset and has

no spatial-temporal information. GPCA results in poor performance on this dataset. (3) The performance of GPLVM is similar to PCA. (4) Both KPCA and GPLVM adopt the kernel trick to handle the nonlinear data, but it is challenging to choose an appropriate kernel for a given data. (5) CopulaPCA does not perform very well possibly because the gross errors have been removed in the data preprocessing procedure.

4.3. The estimated precision matrix is informative

We have shown the effectiveness of MN-PCA in terms of dimension reduction and low-rank representation. Furthermore, we take the METABRIC data as an illustrating example to show that the estimated precision matrices also reveal some inspiring patterns in the noise. It contains a large cohort of around 2000 breast cancer patients with detailed clinical measurements and genome-wide molecular profiles. Here we used the famous PAM50 subtypes [40] as the reference classes. To facilitate a fair comparison, we choose the L_1 -norm parameter for QUIC (Fig. 6, left) such that the estimated precision matrix has the same number of edges approximately as that of MN-PCA (Fig. 6, right). Within-subtype interactions detected by MN-PCA, indicated by dots highlighted with red squares along the diagonals, show a higher density compared to QUIC, with biologically relevant subtypes demonstrating a greater number of interactions. To statistically validate these observations, we performed a Chi-square test, yielding a p -value of 0.0997. In addition, MN-PCA inferred precision matrix tends to have more interactions within the same class (Her2 and LumA for example), which is more biologically plausible.

The next example is about 3sources data, which is collected from three well-known online news sources, including BBC, Reuters, and The Guardian. Here we use the documents collected from the BBC as an example. The document term matrix is of size 352×3560 . We remove the terms that appear less than 20 times in documents. There are 700 terms left. We focus on the estimated precision matrix in the feature space, which reveals the relationship of terms in the documents. The estimated rank $k = 7$, and there are 651 edges. We note that semantically related terms tend to have interactions. For example, the top 5 terms (chosen by absolute values) pairs are “premier/league”, “study/research”, “executive/chief”, “minister/primer”, and “journal/study”, respectively. Those pair of terms frequently appear in the same documents.

5. Discussion and conclusion

We propose MN-PCA to model the graphical noise in both the feature and sample spaces. MN-PCA can be regarded as the minimization of the reconstruction error over the generalized Mahalanobis distance. PCA is a special case of our model. We develop two algorithms for inference. The first one is to maximize the regularized likelihood that works well on the synthetic data when the effect of the matrix normal noise is mild and can handle relatively large data. But it is

Table 8

Performance of dimension reduction on the real-world data sets.

Data set	PCA	GPLVM	CoupluPCA	KPCA	GPCA	MN-PCA	MN-PCA-w2
balance-scale	66.37 (0.76)	68.05 (0.74)	67.41 (0.28)	52.51 (0.74)	77.10 (1.13)	77.39 (1.11)	75.30 (0.37)
german	71.02 (0.65)	70.31 (0.36)	70.21 (0.23)	71.05 (0.74)	70.16 (0.08)	71.48 (0.65)	71.88 (0.54)
glass	56.18 (0.94)	56.32 (1.39)	48.73 (2.39)	51.54 (2.03)	49.55 (1.93)	61.01 (2.18)	47.15 (0.89)
heart-statlog	80.33 (0.95)	82.96 (0.65)	83.30 (0.59)	83.52 (0.94)	83.96 (0.74)	82.07 (0.66)	82.70 (0.72)
iono	93.16 (0.67)	93.53 (0.36)	81.06 (1.11)	92.99 (0.57)	92.11 (0.58)	91.17 (0.43)	92.09 (0.67)
sonar	85.83 (2.21)	71.16 (2.49)	65.30 (2.14)	85.54 (2.34)	81.92 (1.29)	85.34 (1.68)	81.15 (2.15)
tae	43.13 (4.47)	42.71 (5.00)	32.74 (2.33)	40.71 (2.77)	41.73 (2.21)	49.55 (3.11)	42.43 (4.35)
vehicle	45.72 (0.64)	46.60 (0.71)	38.62 (0.98)	43.83 (0.89)	35.83 (1.13)	51.72 (0.93)	58.17 (0.93)
wisc	96.85 (0.13)	96.81 (0.14)	96.75 (0.07)	96.87 (0.17)	96.61 (0.10)	96.50 (0.24)	96.15 (0.17)
zoo	89.51 (1.25)	85.72 (1.38)	81.08 (1.23)	87.50 (1.95)	89.29 (1.40)	76.45 (4.61)	90.46 (5.21)
metabric	66.12 (0.28)	65.85 (0.32)	65.33 (0.32)	67.08 (0.32)	67.12 (0.43)	67.74 (0.62)	61.25 (0.34)
gse29505	92.86 (0.56)	92.86 (0.56)	88.31 (0.70)	92.59 (0.55)	93.62 (0.68)	93.24 (0.52)	92.31 (0.61)
3sources	70.55 (1.82)	69.89 (0.99)	59.16 (0.80)	70.59 (1.22)	47.96 (2.23)	76.78 (0.88)	63.15 (1.26)
gesture	54.49 (0.16)	54.96 (0.47)	52.46 (0.36)	54.82 (0.32)	57.06 (0.28)	55.42 (0.32)	57.72 (0.46)
eeg-eye	61.94 (0.27)	62.22 (0.30)	65.81 (0.27)	65.93 (0.18)	65.90 (0.18)	64.25 (0.44)	65.54 (0.30)
mfeat	90.53 (0.28)	91.55 (0.21)	70.85 (0.56)	88.71 (0.35)	76.13 (0.34)	90.71 (0.41)	69.54 (1.04)
radar	95.57 (0.23)	96.17 (0.20)	77.89 (0.34)	95.47 (0.19)	96.71 (0.30)	95.84 (0.13)	89.80 (0.32)

Top two of the cross-validated accuracies are shown in bold. GPCA (SS) and KPCA with linear kernel are shown. Due to space limitations, the results of PPCA, KPCA with different kernels, GPCA(LS, LL, SL) are shown in the Supplementary Materials.

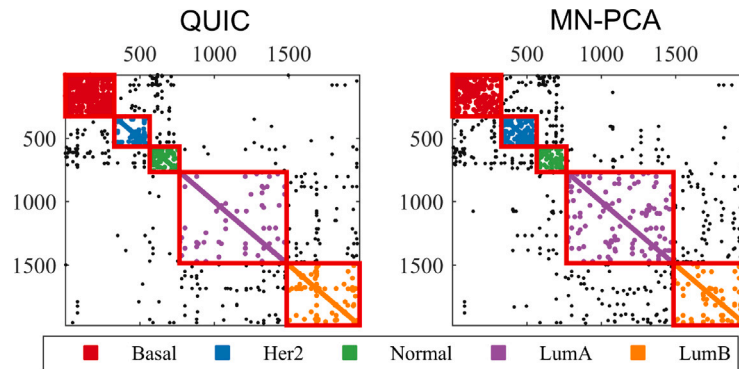


Fig. 6. Visualization of the MN-PCA estimated precision matrix in the sample space, reordered by subtypes ($k = 5$). Displaying the top 1000 interactions, the within-subtype interactions are highlighted within red squares. The MN-PCA precision matrix (159) suggests a propensity for more within-subtype connections compared to QUIC (133), with a p -value of 0.0997 from the Chi-square test.

not robust when the condition numbers get larger. To address this challenge, we propose to minimize the Wasserstein distance between the transformed residual and the white noise. It is more robust when the effect of the matrix normal noise is substantial. Extensive numerical results show that considering the structural noise brings an improvement in classification, suggesting our methods find better low-rank representations. Moreover, the inferred precision matrices are informative and can help us understand the underlying structure of the noise. MN-PCA is well-suited for scenarios where a dual consideration of both sample and feature spaces is essential. Examples include gene expression data analysis demonstrates its efficiency in revealing underlying patterns in biological conditions (samples), as well as corpus data analysis, enabling a thorough exploration of the relationship in documents (samples) and words (features). Additionally, it addresses

challenges commonly encountered in real-world applications, offering a robust solution for datasets with intricate dependencies among samples and features, respectively.

Several questions remain to be investigated. First, the estimation of the precision matrices is computationally expensive, which makes the proposed algorithms inefficient for large data. It is worth studying how to develop a more efficient and robust algorithm for big data. Despite the fact that the estimation of the precision matrices is empirically reasonable, the identifiability of the matrix normal noise needs to be further investigated. Second, our proposed methods are not guaranteed to provide a better low-rank representation for clustering and classification. As we demonstrate in the real-world experiments, the estimated precision matrices are also informative. If the inferred structure of the noise captures the information related to clustering and classification,

the performance of the low-rank representation possibly decreases. One crucial problem is what is the noise, and what is the signal? Conventional methods such as PCA assume that the noise is white noise, which is seldom satisfied. In this paper, we model the graphical noise by matrix normal distribution. Although it is difficult to determine what information is captured in the estimated noise without external knowledge, our methods provide an approach for data exploration for users to obtain the low-rank representation and discover the underlying structure of the noise simultaneously.

MN-PCA can be extended from two aspects: (1) it can be extended to matrix variate data, where one has multiple matrix variate observations. (2) it is worth generalizing MN-PCA for tensor data by the tensor normal distribution, enabling us to explore the correlation of the noise in the high-order data. It is also worth investigating using minimizing Wasserstein distance for other matrix factorization methods. Specifically, consider the general matrix factorization model with additive noise, i.e., $Y = XW^T + E$, $E \sim P_n$, where P_n indicates the additive noise distribution. Instead of maximizing the likelihood estimation, one can infer X and W by minimizing the Wasserstein distance $W(P_r, P_n)$, where P_r is the empirical distribution of residues $Y - XW^T$.

CRedit authorship contribution statement

Chihao Zhang: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis. **Kuo Gai:** Writing – original draft, Validation, Software, Methodology, Formal analysis. **Shihua Zhang:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declares that there is no conflict of interest regarding the publication of this article.

Data availability

All the used data is publicly available.

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Appendix A. Supplementary data

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